THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of radius 14 cm and angle 90° is: 1

(a) 154 cm² **(b)** 616 cm² **(c)** 77 cm² **(d)** 308 cm²

2. The arc length of a sector of radius 7 cm and angle 60° is: 1

(a) $\frac{22}{3}$ cm **(b)** 22 cm **(c)** $\frac{44}{3}$ cm **(d)** 11 cm

3. If the area of a circle is 616 cm², its radius is: 1

(a) 28 cm **(b)** 21 cm **(c)** 7 cm **(d)** 14 cm

4. A student states: "A 360° sector of a circle of radius 7 cm has area 44 cm²." The correct response is: 1

(a) Incorrect — 44 cm is the circumference; area is 154 cm²
(b) Area is 22 cm²
(c) Correct
(d) Cannot be determined

5. For the angle swept by the hour hand in 2 hours, Student A says 60°; Student B says 30°. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The volume of a cylinder of radius 7 cm and height 2 cm is: 1

(a) 154 cm³ **(b)** 308 cm³ **(c)** 616 cm³ **(d)** 88 cm³

7. The curved surface area of a hemisphere of radius 14 cm is: 1

(a) 1232 cm² **(b)** 2464 cm² **(c)** 616 cm² **(d)** 1848 cm²

8. A cone and a cylinder have the same base and height. The ratio of their volumes is: 1

(a) 3 : 1 **(b)** 1 : 3 **(c)** 2 : 3 **(d)** 1 : 2

9. A student states: "A sharpened pencil is a cylinder joined with a hemisphere." The correct response is: 1

(a) It is a cuboid
(b) Correct
(c) Incorrect — it is a cylinder joined with a cone
(d) Cannot be determined

10. Two cubes of edge a are joined. Student A says the surface area is $10a^2$; Student B says $12a^2$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of radius 21 cm with angle 45°. 2

12. Find the length of an arc subtending 36° in a circle of radius 10 cm. (Use $\pi = 3.14$) 2

13. A cubical box of edge 8 cm has a hemisphere of diameter 8 cm on its top face. Find the total surface area. 2

14. A cone of radius 3 cm and height 4 cm stands on a cylinder of the same radius and height 10 cm. Find the volume. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the minor segment of a circle of radius 12 cm when the sector angle is 90°. (Use $\pi = 3.14$) 3

16. A solid is a cylinder of diameter 14 cm and height 15 cm with a hemisphere attached to one end. Find its total surface area. 3

17. A sector of a circle of radius 15 cm has angle 120°. Find its area, arc length and perimeter. (Use $\pi = 3.14$) 3

OR

A clock's minute hand is 10.5 cm long. Find the area swept by it from 9:00 to 9:40. 3

18. A tent is a cylinder of height 2.5 m and radius 7 m surmounted by a cone of slant height 8 m. Find the area of canvas used. 3

OR

A hemispherical bowl of internal radius 9 cm is full of milk. Find the volume of milk in litres. (Use $\pi = \frac{22}{7}$) 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Sector OAB of a circle has radius 21 cm and angle 60°; AB is the chord. Find 5

(i) the arc length

(ii) the sector area

(iii) the segment area

(iv) the perimeter of the segment. (Use $\sqrt{3} = 1.73$)

OR

A car has two non-overlapping wipers, each of length 21 cm sweeping 120°. 5

(i) Find the total area cleaned per sweep.

(ii) Find the area if one wiper breaks.

(iii) What percentage of a 5000 cm² windshield is cleaned per sweep?

20. **Case study:** A grain silo is a cylinder of radius 3 m and height 10 m topped by a cone of height 4 m. (Use $\pi = 3.14$) 5

(i) Find the volume of the cylinder.

(ii) Find the volume of the cone.

(iii) Find the total capacity.

(iv) Find the sheet area for the curved surfaces.

OR

A building is a cylinder of radius 3.5 m and height 6 m with a hemispherical dome. 5

(i) Find the volume of air inside.

(ii) Find the inner surface area of walls and dome.

(iii) Find the cost of painting it at ₹10 per m².

END OF QUESTION PAPER — **Total: 40 Marks**